

JAKE WALKLATE

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SUMMARY

Penultimate multidisciplinary engineering student with 10 years of work experience including delivering production systems in early-stage SaaS startups, independent R&D environments and customer focused experiences. Proven ability to take products from concept to launch and work in teams receiving feedback. Specialist in high-stakes hardware-software convergence, from bare metal kernel-level troubleshooting to cross-platform cloud-based healthcare architectures.

EDUCATION

B Eng (Hons) in Artificial Intelligence, Mechatronics & Biomed Engineering | **UTS** Expected 2027

- Summary coursework: CNN for Biometric ID (TensorFlow/Python), Bio-compatible 3D Printed Polymers, FPGA Circuits and STM32 Microcontrollers, Sensors for Robotics.

B Med Sci in Pathology | **UTS** Expected 2027

- Summary coursework: Genetics and Recombinant Molecular Engineering, Organic Chemistry, Bioinformatic Data Processing.

Cert IV in Programming & Cyber Security | **TAFE** Completed 2020

- Summary coursework: RDBMS with SQL, Automated Scripting, Object Oriented Programming, Automated Software Testing, RESTful APIs, Virtualisation of Linux, Secure Software Practices & Vulnerabilities.

EXPERIENCE

Application Developer | **Medicamina**

- Led an entrepreneurial R&D project focused on genetic clinical support and patient care using Flutter and Material Design, selecting tools based on accessibility and cross-platform compilation targets.
- Addressed critical pharmacogenetic gaps where ~50% of SSRIs were ineffective by developing intelligent technical decision support tools and trained a neural network for blood classification.
- Developed and maintained authorization-based CRUD data APIs using Microsoft Azure and PostgreSQL.

Founding Engineer | **Coreable**

- Architected and deployed production web servers – starting from an incubator with 14% equity – collaborating closely with UI/UX support and stakeholders in an Agile method.
- Built cross-platform front ends (mobile/desktop) with React (TypeScript/Bootstrap), non-blocking backends with GraphQL Apollo Server/Node.JS/MySQL and maintained isolated test environments with Docker.
- Managed system updates, code reviews (with git version control), and CI/CD pipelines with GCP tooling and remote cloud configuration via BASH terminal and SSH.
- Winners of the UTS' best incubator product award for addressing critical modern hiring practices.

Programming Analyst | **ClockOn**

- Acted as a Technical Lead for an enterprise-wide application rewrite from legacy Delphi to modern HTML/C# ASP.NET stack, making decisions enabling accessibility and cross browser support.
- Developed enterprise mobile/web solutions using JavaScript and third-party UI components.

FURTHER BACKGROUND

Extensive tenure in Hospitality (McDonalds, ConceptAte) and Construction (Labouring, Fiber Optic Cabling). This time of diverse manual labour and customer-facing roles developed grit, store management experience, and a pragmatic mindset that informs my work ethic today.

- Awarded most liked peer at ConceptAte by management and staff.

KEY PROJECTS

OpenBio

- Created a Tauri desktop app with a React+Tailwind built user interface to make it easy to store and manage critical and sensitive laboratory data in a unified system.
- Automated software packaging installation and management to minimise friction between biological experimentation and bioinformatic tool usage.
- Utilised Direct Memory Access to reduce CPU load when moving large files from disk to memory and implemented mDNS for easy local connectivity.

International Space Station Orbit Tracker

- Used Autodesk Fusion (CAD) to draw a 3D printable model (PLA) for the purpose of housing an Arduino based IoT project which tracks the orbit of celestial bodies in LEO using the SGP4 calculation process.
- Selected appropriate polymer materials (PA-6) and greases for bearings to allow the lowest friction and wear in moving parts, based on materials' toughness, strength and coefficient of friction.
- Called external APIs via HTTPS (C++) to find ISS TLE data and the earth's magnetic declination for device configurations, while implementing hard-coded backups in case of network failure.
- Utilised PWM for stepper motor control while incorporating independent azimuth position feedback via GPS and magnetometers for precise calibrations.

Arduino Community Edition IntelliSense

- Published a free vite-coded Visual Studio Code plugin to fix a bug in the Visual Studio Code Arduino plugin where IntelliSense wouldn't resolve type definitions or functions.
- Received 500+ downloads and 5-star reviews.
- Optimised the plugin to use in memory cache and debounced compilation to minimise CPU and battery usage.

PUBLICATIONS / TECHNICAL CASE STUDIES

The Pharmacology of Silence

<https://svnty.is-a.dev/book/the-pharmacology-of-silence>

- A book on a first-hand and critical analysis of the Australian mental health system – examining forced medication practices, diagnostic inaccuracy, institutional accountability, and patient rights. Covers the pharmacogenomic mechanisms of paliperidone, zuclopenthixol and venlafaxine.

Thanks a lot Claude

<https://svnty.is-a.dev/blog/thanks-a-lot-claude>

- Performed a root-cause analysis of a bricked ARMv7 FDM printer, diagnosing an unescaped character bug in a cloud JSON-RPC API after flashing the eMMC and performing a log analysis, and then successfully ported the 1990's FPS shooter DOOM via Docker – restoring 110% of the 3D printer's functionality.

ATTENDED WORKSHOPS

Brain and Computer Interface with Google TensorFlow | AntNeuro

- Hands-on workshop on how to bridge the gap between Neuroscience and Computational Intelligence to design and develop Brain-Computer Interface.

CORE COMPETENCIES

Software Development: Flutter, TypeScript, Next.js, Angular, C#, Java, SQL, Docker, Rust.

Medical: Pharmacogenetics, Molecular Biology, 3D Bioprinting, BLAST.

Product Design: Arduino, STM32, FPGA, SolidWorks, Fusion, FDM printing.

Intelligent Systems: Neural Networks, Decision Trees, Agile/Scrum, Waterfall, UML, Fuzzy Logic.

REFERENCES